SREDAY SAN FRANCISCO

When the On-Call Engineer Is an Agent

Incident Response for AI-Driven Systems

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The Pager Went Off. No One Paged It.

An agent took an action nobody approved.

No error was thrown. No alert fired. No human was in the loop when it happened — only when someone noticed the outcome.

This is what failure looks like now.

SRE Was Built for a Different Kind of Failure

Traditional Systems

- Deterministic behavior
- Clear, catchable error states
- Stable over time
- Decisions traceable via stack traces

AI Agents

- Probabilistic behavior
- Fail silently, no clean error
- Drift as underlying models update
- Decisions hard to reconstruct after the fact

What Breaks First in Production

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Unplanned tool calls

The agent invokes an action it wasn't explicitly told to take.

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Silent partial failures

The task looks complete. The output is wrong or incomplete.

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Behavioral drift

Same prompt, different behavior after a model update underneath it.

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Cascading agent-to-agent errors

One agent's mistake becomes the next agent's input.

The Ownership Gap

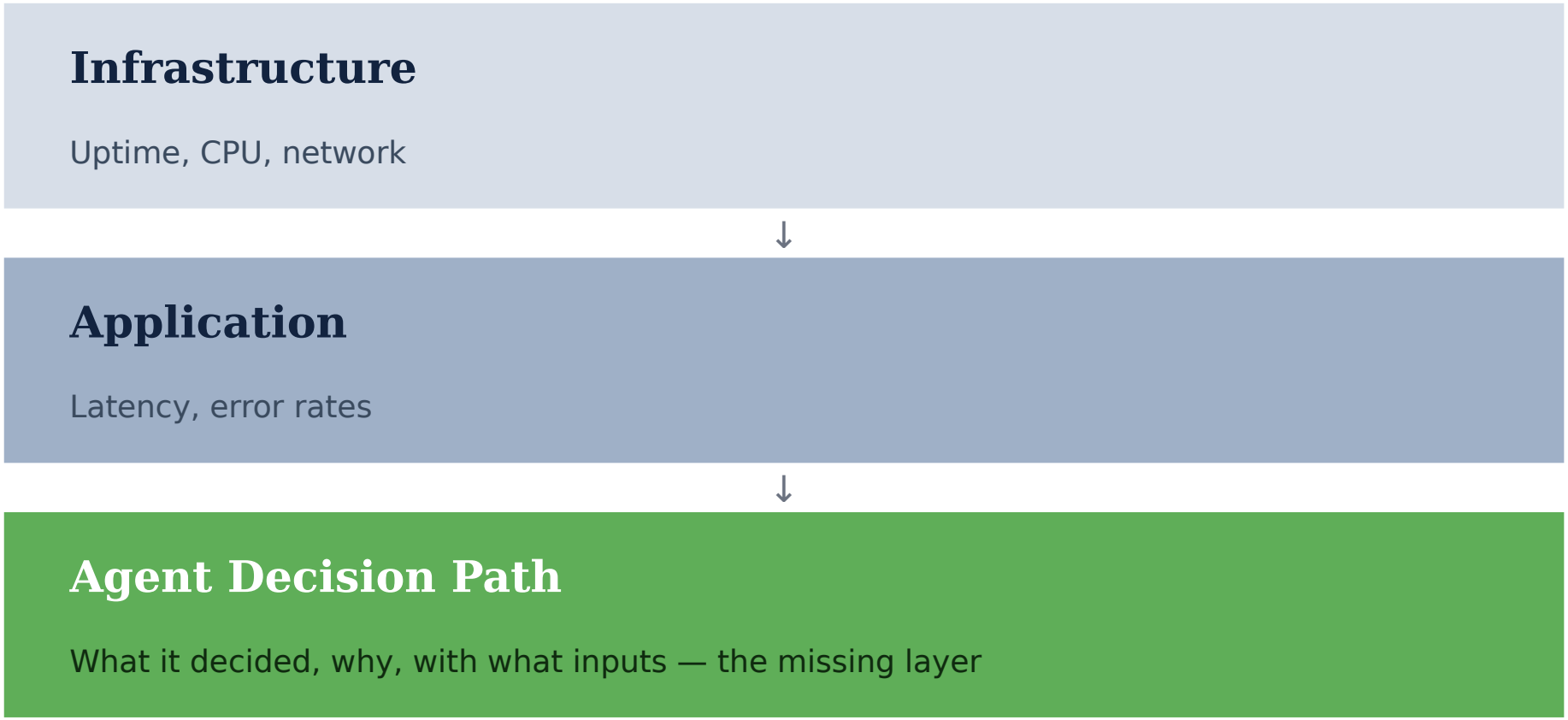
“Who gets paged when the agent decided wrong?”

- No incident classification for “the agent did something wrong.”
- No on-call rotation built for model behavior, not just infrastructure.
- No postmortem process designed for probabilistic systems.
- No clear owner when an agent's action costs money, time, or trust.

A New Failure Taxonomy

Classic Incident Category	Agent-Era Addition
Infrastructure outage	Model or behavioral drift
Code bug	Prompt or tool misuse
Data corruption	Hallucinated output propagated downstream
Capacity limit exceeded	Autonomous action outside intended scope

Observability Has to Move Up the Stack



Incident Classification for Agent Failures

SEV 1	Agent took an irreversible action outside approved scope
SEV 2	Agent produced incorrect output that reached a customer or downstream system undetected
SEV 3	Agent behavior drifted but was caught before impact

Redesigning On-Call for Agentic Systems



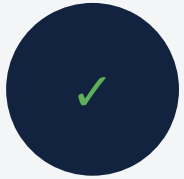
Traditional On-Call Engineer

Owns infrastructure, latency, and system-level failure as before.



Model / Behavior Owner

Understands the model's decision patterns and can assess whether behavior is expected or drifted.



Human Reviewer

Approves or halts high-risk autonomous actions before they execute, where the blast radius warrants it.

Postmortems for Probabilistic Systems

“It won't happen again” doesn't work when behavior is probabilistic.

What was the agent's decision path, step by step?

Could this recur with different inputs, not just the exact same ones?

What guardrail catches this class of failure — not just this one instance?

Where to Start This Quarter

1. Define failure modes specific to your agents — don't reuse the infra list.
2. Add decision-path logging before you need it, not after an incident.
3. Assign clear ownership before an agent-caused incident forces the question.
4. Run a tabletop exercise for an agent-caused incident this quarter.

The Gap Doesn't Stay Invisible for Long

The companies moving fastest on AI adoption are often moving slowest on AI reliability.

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